# M Minimum Records

Time limit: 1s

VegiCat has $n$ closed integer intervals $[l_1, r_1], [l_2, r_2], \ldots, [l_n, r_n]$.



He wants to construct a sequence $a_1, a_2, \ldots, a_n$ by choosing an integer $a_i$ from each interval $[l_i, r_i]$. That is, $l_i \le a_i \le r_i$ for every $1 \le i \le n$.

A position $i$ is called a **record position** if $a_i > a_j$ for every $1 \le j < i$. In particular, position 1 is always a record position.

Help VegiCat minimize the number of record positions. Output this minimum number.

## Input

The first line contains an integer $T$ ($1 \le T \le 10^4$), the number of test cases. The description of the test cases follows.

The first line of each test case contains an integer $n$ ($1 \le n \le 10^6$), the number of intervals.

The $i$-th of the next $n$ lines contains two integers $l_i, r_i$ ($-10^9 \le l_i \le r_i \le 10^9$), describing the interval available at position $i$.

It is guaranteed that the sum of $n$ over all test cases does not exceed $10^6$.

## Output

For each test case, output an integer — the minimum possible number of record positions.

| Sample Input 1 | Sample Output 1 |
|---|---|
| 2<br>4<br>1 1<br>1 100<br>20 20<br>30 30<br>8<br>-3 -1<br>-5 4<br>0 1<br>2 2<br>1 8<br>5 6<br>10 12<br>12 12 | 2<br>4 |

## Note

In the first test case, one optimal sequence is $a = [1, 100, 20, 30]$. Its record positions are 1 and 2.

In the second test case, one optimal sequence is $a = [-1, 4, 1, 2, 8, 6, 12, 12]$. Its record positions are 1, 2, 5, and 7. Position 8 is not a record because its value equals the maximum seen so far.